

29.

Ans: C

For the electron,

Gain in KE = Loss in EPE

$$\frac{1}{2} m_e v_e^2 = eV$$

$$\frac{1}{2} (9.11 \times 10^{-31}) v_e^2 = (1.6 \times 10^{-19})(1200)$$

$$v_e = 2.05 \times 10^7 \text{ ms}^{-1}$$

Similarly for the proton, $v_p = 4.80 \times 10^5 \text{ ms}^{-1}$

$$\frac{\lambda_e}{\lambda_p} = \frac{h/m_e v_e}{h/m_p v_p} = \frac{m_p v_p}{m_e v_e} = 43$$

OR

Gain in KE = Loss in EPE

$$\frac{p^2}{2m} = eV$$

$$p = \sqrt{2meV}$$

$$\frac{\lambda_e}{\lambda_p} = \frac{h/p_e}{h/p_p} = \frac{p_p}{p_e} = \frac{\sqrt{2m_p eV}}{\sqrt{2m_e eV}} = \frac{\sqrt{m_p}}{\sqrt{m_e}} = 43$$

30.

Ans: D

Increasing the p.d. will result in a smaller minimum wavelength as the electrons will have